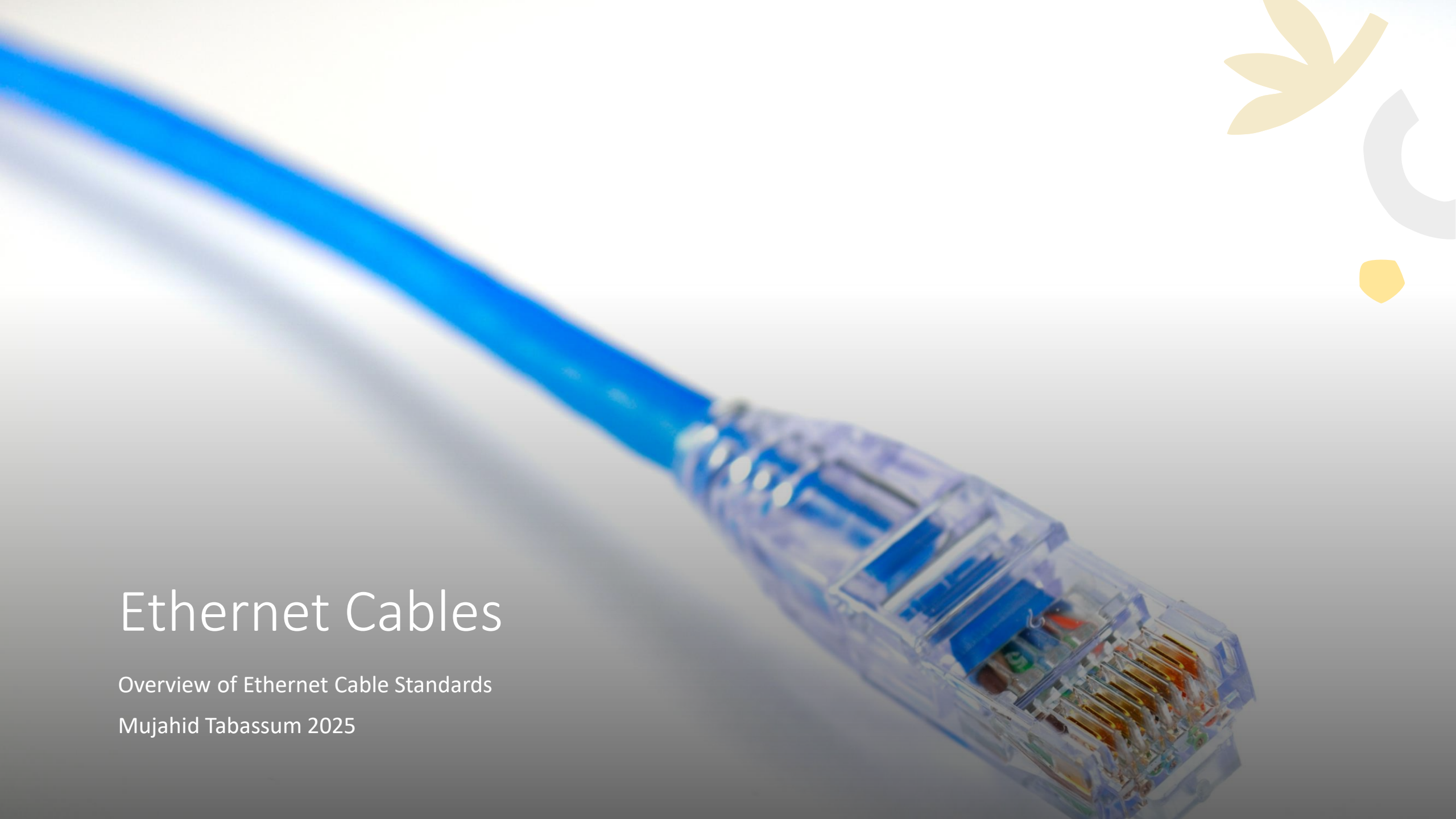




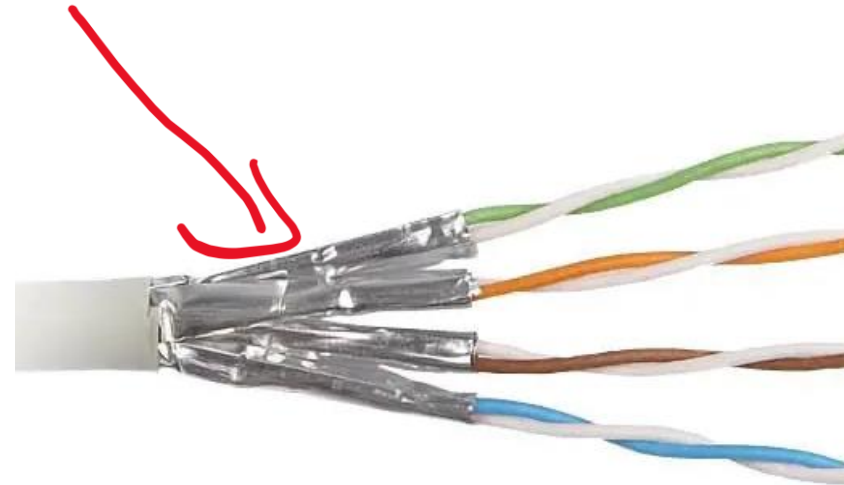
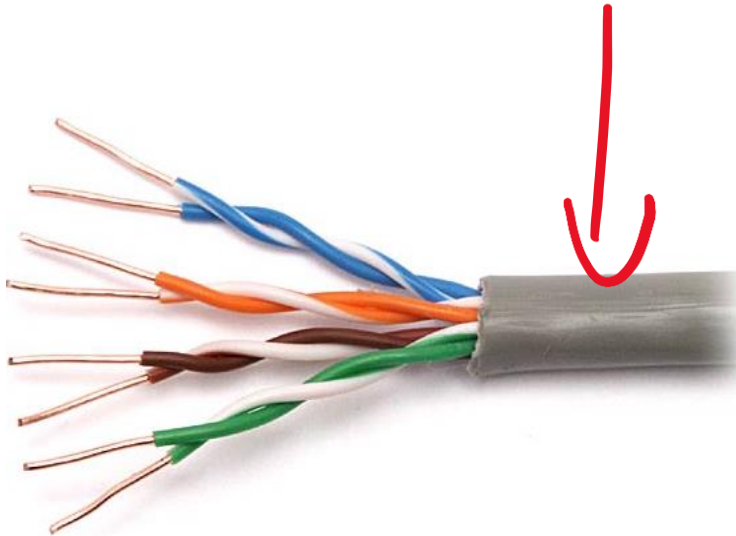
Ethernet Cables

Overview of Ethernet Cable Standards

Mujahid Tabassum 2025

Introduction

- Ethernet cables connect devices within local networks.
- Developed over the years to support higher speeds & reduce interference.
- Classified by "Category" and Construction Type
 - (e.g. Unshielded Twisted Pair, Shielded Twisted Pair/Foil Twisted Pair).

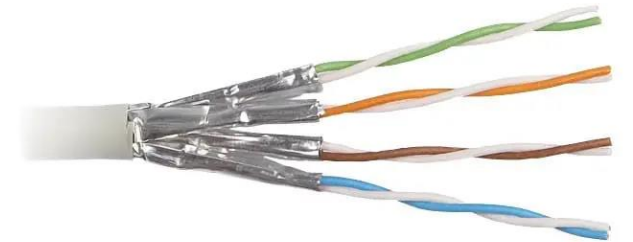
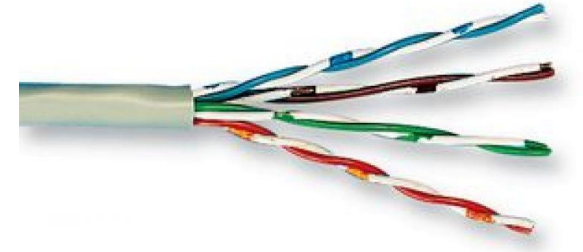


Main Ethernet Categories

- Cat5e
 - Speeds up to 1 Gbps.
 - Enhanced to reduce crosstalk
- Cat6
 - 1 Gbps possible for distances up to 100 meters. 10 Gbps possible under 55 meters.
 - Better performance and reduced crosstalk compared to Cat 5e.
 - Cat6a (augmented) offers 10 Gbps up to 100 meters)
- Cat7
 - Speeds up to 10 Gbps. Uses shielded twisted pairs for less crosstalk. Stricter standards for shielding
- Cat8
 - Designed for Data Centres where 25-40Gbps is required
 - Shielded

Construction

- UTP (Unshielded Twisted Pair)
 - The most common type used in most installations.
- STP (Shielded Twisted Pair)
 - Has shielding to prevent external interference.



Connectors

- RJ45 connectors used on cables
- RJ45 sockets used on devices/walls
 - E.g. switches



Poorly terminated UTP cable

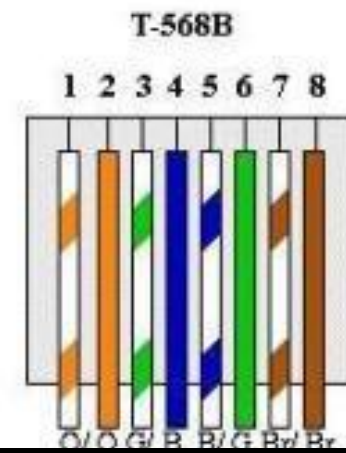
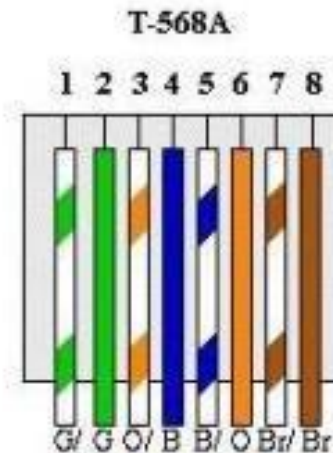
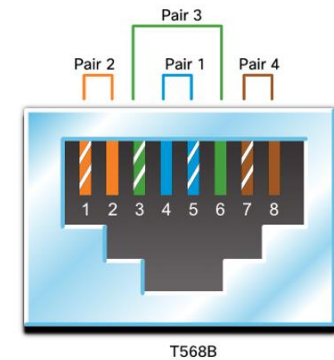
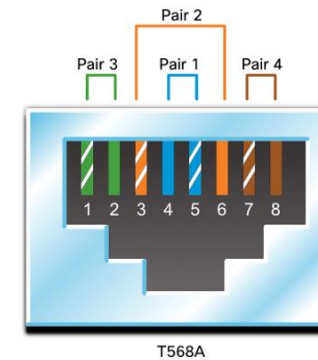


Properly terminated UTP cable



Wiring Standards

- 8 wires, how/what order do you connect them?
- The two most common standards for Ethernet cable wiring are T568A and T568B.
- Once the same standard is used at each end, there should be no problem.



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